Practical Exam Presentation

Product Sales

Pens and Printers

December 4th, 2025

Background + Business Goals

- *Pens & Printers* serves to provide a supply of high quality office products to enable customers to be more creative & focused on tools for brainstorming.
- As the ways in which consumers buy products is changing, sales tactics have to change too. Launching a new product line is expensive & the sales team needs to make sure they are using the best techniques to sell the new products effectively. The best approach may vary for each new product so the team needs to learn quickly what works & what doesn't.
- Three different sales strategies were tested for these purposes.
    - **Email**: Customers in this group received an email when the product line was launched, & a further email three weeks later. This required very little work for the team.
    - **Call**: Customers in this group were called by a member of the sales team. On average members of the team were on the phone for around thirty minutes per customer.
    - **Email & Call**: Customers in this group were first sent the product information email, then called a week later by the sales team to talk about their needs & how this new product may support their work. The email required little work from the team, the call was around ten minutes per customer.

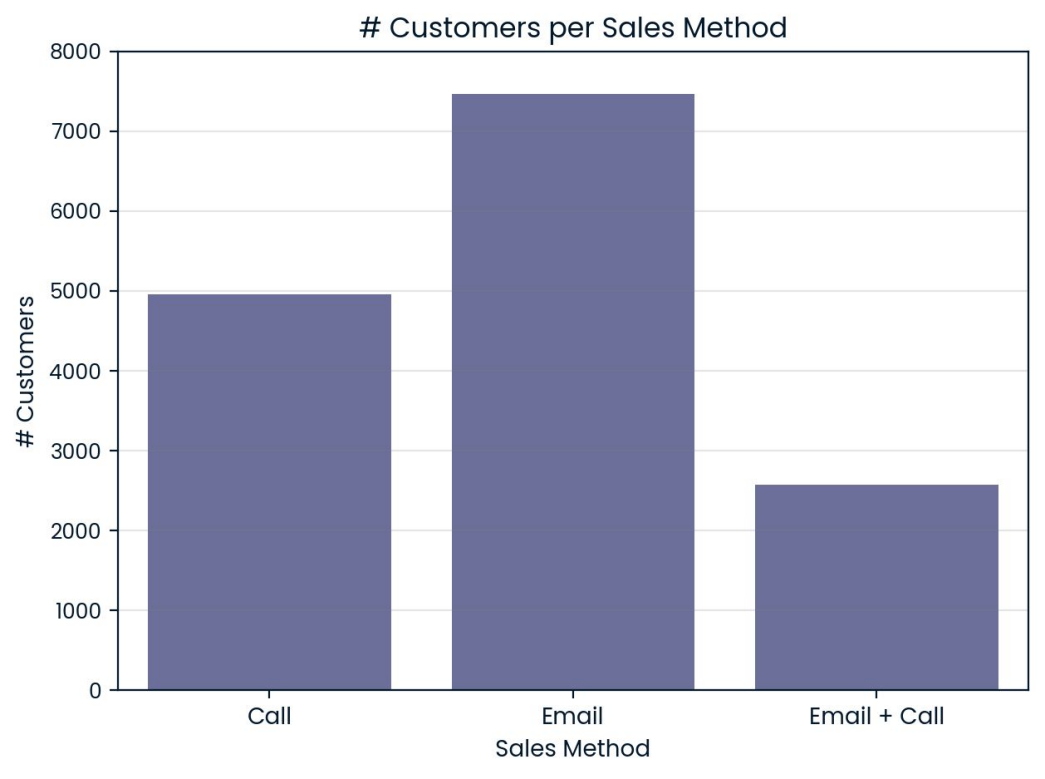
Background + Business Goals

- In anticipation of a meeting with the executive team to update them on sales approaches for the new product line, the sales team want to know the following.
  - How many customers were there for each approach?
  - What does the spread of the revenue look like overall? And for each method?
  - Was there any difference in revenue over time for each of the methods?
  - Are there any other differences between the customers in each group?
  - Which method(s) should the sales team continue to use given that they take differing amounts of time from the team.

Outcomes

Differences in the sales between sales approaches - Number of customers

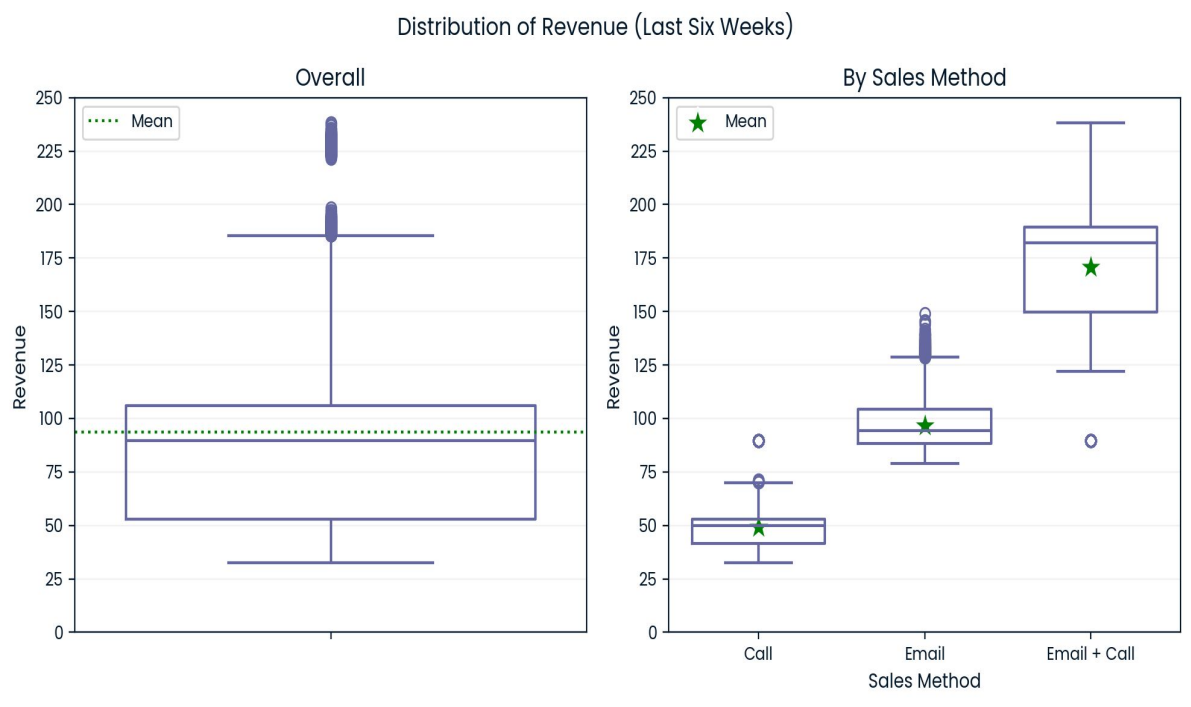
- Of the 15,000 customers who have made purchases from the new product line in the last six weeks, about half of them only received an email, about a third only received a phone call, & about a sixth received both.
- In other words, for every six customers, generally three of them were contacted via email, two via phone, & one via both.



Outcomes

Differences in the sales between sales approaches - Revenue

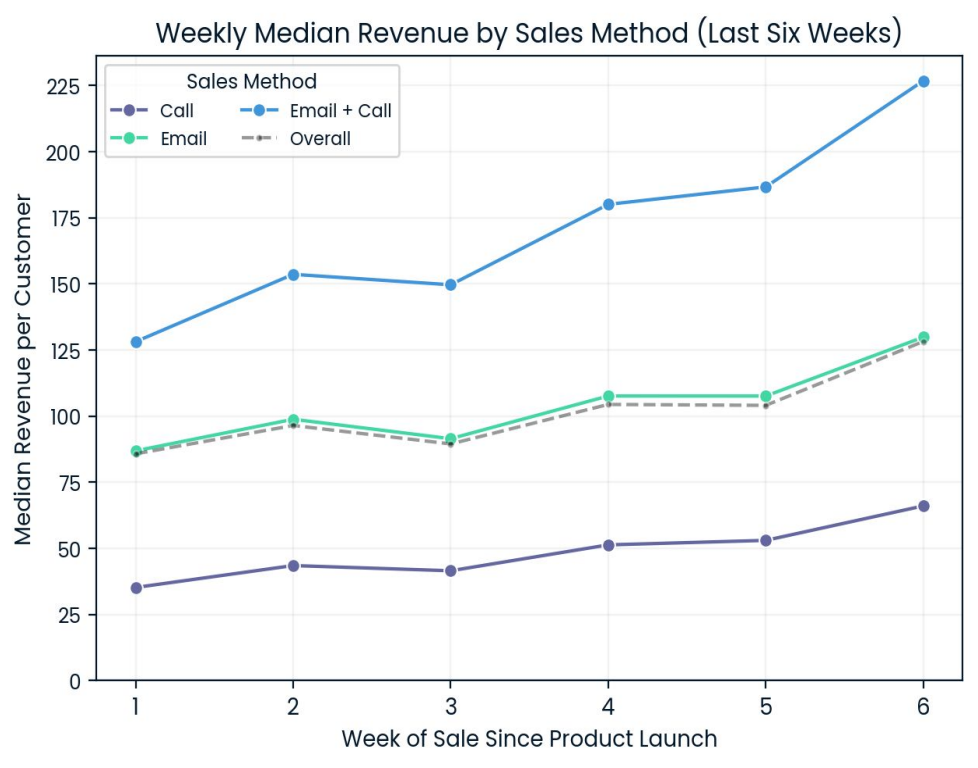
- Revenue generated by the "Call" group was generally the least of the three methods by a considerable margin.
- Emailing people generally resulted in a greater revenue per customer than phoning them, but this wasn't the most effective method.
- When contacted by both email & phone, customers appeared significantly more likely to spend more on the new products.



Outcomes

Differences in the sales between sales approaches - Revenue Over Time

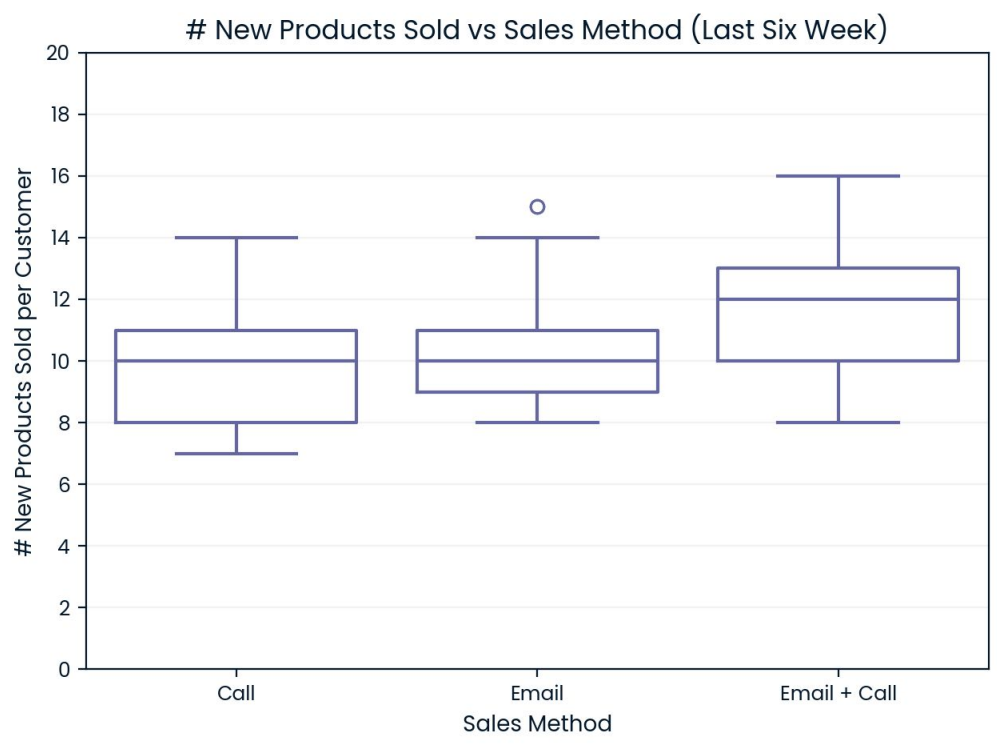
- Over the last six weeks, all three sales methods saw a considerable increase in typical revenue per customer. These increases suggest that all three sales techniques became increasingly more effective over time in terms of generating revenue.
    - For customers contacted via phone only, the typical weekly revenue increased by about 89%.
    - For customers contacted via email only, the typical weekly revenue increased by about 49%.
    - For customers contacted via email & phone, the typical weekly revenue increased by about 77%.
- Multiple factors could be at play regarding these trends like the quality of advertising, the number of customers who were contacted, or other variables.



Outcomes

Differences in the sales between sales approaches - Other Differences

- Additional variables were analyzed according to the sales method utilized. Of them, only the number of products purchased appeared to have a meaningful relationship with the sales method used.
- More specifically, customers who were contacted via email <u>&</u> phone typically bought two more products (in quantity) than a customer who was contacted via email <u>*or*</u> phone.



Business Metrics

To help quantify the general effectiveness of each sales method, a revenue efficiency metric was built that is based on the revenue generated & time spent advertising to a typical customer. For example, the typical revenue per customer in the "Email" group divided by the typical time spent advertising to these customers gives an estimate of the efficiency regarding how much revenue is generated over a period of time that is spent advertising via email (e.g. an hour).

- This metric was chosen because of what variables were available, & because a business' revenue & their profits are some of the most indicative metrics of success.

The average time spent contacting customers for each sales approach are as follows.

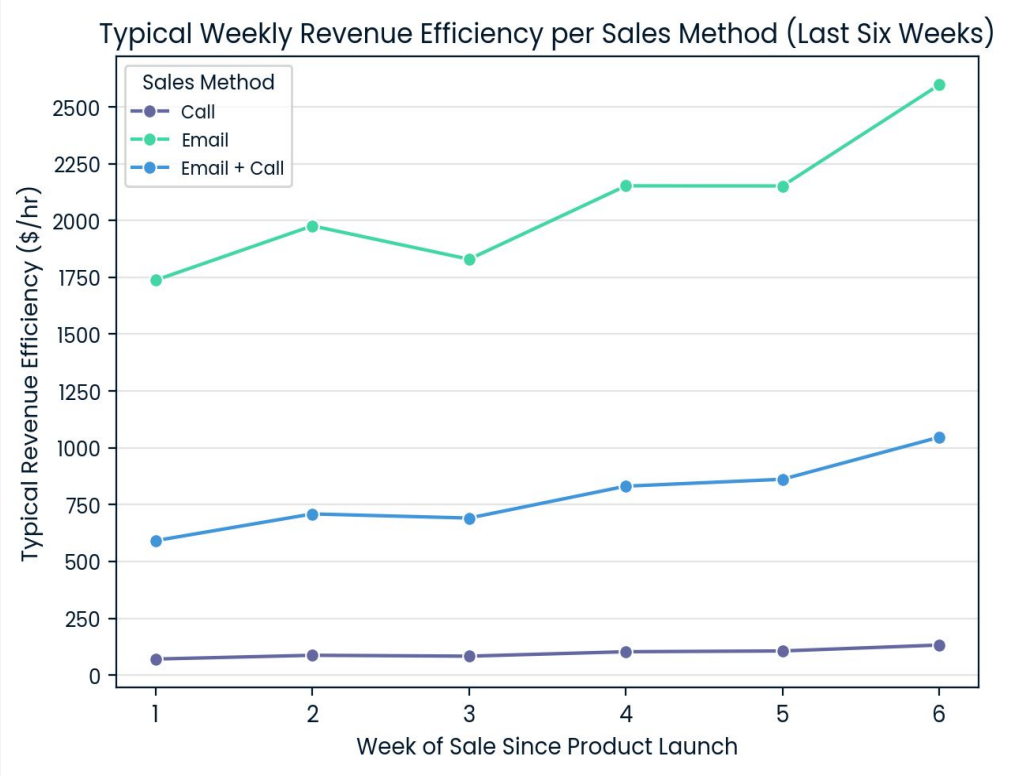
- **Call**: Team members were on the phone for around 30 minutes on average per customer.
- **Email**: It is estimated that constructing & sending an email took no more than three minutes on average.
- **Email + Call**: Customers were sent an email & called about a week later. These phone calls were around ten minutes on average per customer. Adding the email totals about 13 minutes for this process per customer.

Business Metrics

Generally, the value of this metric increased over the last six weeks. In the most recent week of data, the three sales methods have the following typical revenue efficiencies.

- **Call**: About $132 per hour.
- **Email**: About $2,597 per hour.
- **Email + Call**: About $1,047 per hour.

Not only can this metric assess how the relative effectiveness of each sales method has evolved in the past, but it can also be used to project revenue efficiencies in the near future & provide recommendations regarding how the sales team's time spent advertising can be optimized.



Recommendations

In contextualizing the findings from the revenue efficiency metric with those of previous analyses, it is clear that only calling customers was the least productive sales strategy by far in terms of both outright revenue as well as revenue efficiency. Ideally, this technique, given how much less profitable it has been in comparison to the other two methods, should be utilized at a minimum; however, this is not to say that talking to customers directly over the phone does not have some value.

Moreover, with how much time the sales team typically spends on these phone calls, it would be more productive for them to spend most of their time emailing customers (the “Email” method) & occasionally calling some people as a follow up (the “Email + Call” method).

Utilizing both sales techniques resulted in the most revenue per customer, but this method hasn't been quite as efficient as only emailing people. Therefore, emailing customers should be the primary sales method which hasn't been the case in the last two weeks.

Recommendations

On another note, it is important that the company continues to monitor this metric (& any others they have used or found useful) because customer purchasing habits can change easily. Allowing for flexibility is essential if, for example, customers were to suddenly become considerably more enamored by the products via phone call & less so by email. Moreover, if total revenue were to stop increasing or decline, new strategies or plans may need to be considered or conceived so that the company doesn't become uncompetitive or unprofitable.

- In summary, an optimal strategy moving forward would involve mostly emailing customers & occasionally following up with some of them by phone (the "Email" & "Email + Call" sales methods). Calling people who have not been emailed (the "Call" method) should be used at a minimum.

Recommendations - Optimizing Time

With this information in mind, it would be useful for the sales team to have a more defined plan as to how to optimize their time across the three sales methods. Since emails have the greatest revenue efficiency per customer & they require the least amount of time of the three methods, the revenue would theoretically be maximized if all of the sales team's time was allotted to the “Email” method alone; however, having some diversity in advertising approaches is more ideal to help keep the business competitive, relevant, & flexible.

As such, the balance between the amount of time spent on each sales method depends on how much the company values revenue versus diversity in their advertising approaches. A few examples were created that lay out estimates of revenue & other quantities based on how much of an employee's time per week is spent advertising across each method. These examples all follow the same general strategy discussed previously—most time time is spent sending emails, & little time is spent calling people. Although relying more on emails alone will likely yield more revenue, these examples provide an illustration of how the sales team can fiddle with their time spent advertising to obtain a desirable & satisfactory (theoretical) revenue.

Recommendations - Optimizing Time

From data in sixth week

Sales method	Typical min/cust	Typical hr/cust	rev_eff ($/min)	rev_eff ($/hr)
Email	3	0.05	43.29	2597.4
Email + Call	13	0.216666667	17.445	1046.723
Call	30	0.5	2.202	132.14

Consider 1 team member has 6 hr/day (30 hr/week) to contact customers. Using the sales-method splits in the sixth week:

Sales method	W6 Splits (% customers)	W6 Splits (hrs/week)	Typical min/cust	# cust contacted (wk)	# cust contacted (day)	rev_eff ($/hr)	-->	revenue ($/wk)
Email	0.160423	4.81269	3	96	19	2597.4		$ 12,500.48
Email + Call	0.474756	14.24268	13	65	13	1046.723		$ 14,908.14
Call	0.364821	10.94463	30	21	4	132.14		$ 1,446.22
TOTALS	1	30		182	36			**$ 28,854.85**

Recommendations - Optimizing Time

Tinker with the sales-method splits based off of data from the sixth week

1A

Sales method	Wkly Splits (% customers	Wkly Splits (hrs/week	Typical min/cust	# cust contacted (wk)	# cust contacted (day)	rev_eff ($/hr)	-->	revenue ($/wk)
Email	0.5	15	3	300	60	2597.4		$ 38,961.00
Email + Call	0.45	13.5	13	62	12	1046.723		$ 14,130.76
Call	0.05	1.5	30	3	0	132.14		$ 198.21
TOTALS	1	30		365	72			**$ 53,289.97**

1B

Sales method	Wkly Splits (% customers	Wkly Splits (hrs/week	Typical min/cust	# cust contacted (wk)	# cust contacted (day)	rev_eff ($/hr)	-->	revenue ($/wk)
Email	0.7	21	3	420	84	2597.4		$ 54,545.40
Email + Call	0.25	7.5	13	34	6	1046.723		$ 7,850.42
Call	0.05	1.5	30	3	0	132.14		$ 198.21
TOTALS	1	30		457	90			**$ 62,594.03**

1C

Sales method	Wkly Splits (% customers	Wkly Splits (hrs/week	Typical min/cust	# cust contacted (wk)	# cust contacted (day)	rev_eff ($/hr)	-->	revenue ($/wk)
Email	0.85	25.5	3	510	102	2597.4		$ 66,233.70
Email + Call	0.1	3	13	13	2	1046.723		$ 3,140.17
Call	0.05	1.5	30	3	0	132.14		$ 198.21
TOTALS	1	30		526	104			**$ 69,572.08**